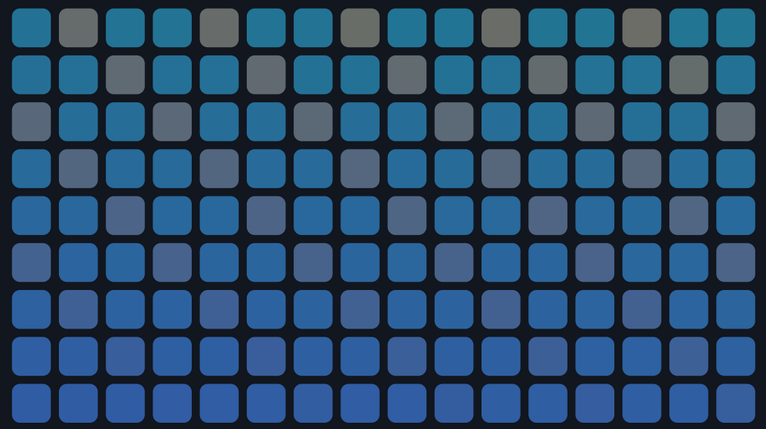
WHY CUDA

Particle life asks the same question thousands of times

That repetition is exactly what GPUs are built for: many small workers doing similar math at the same time.

GPU view

one tiny worker per particle

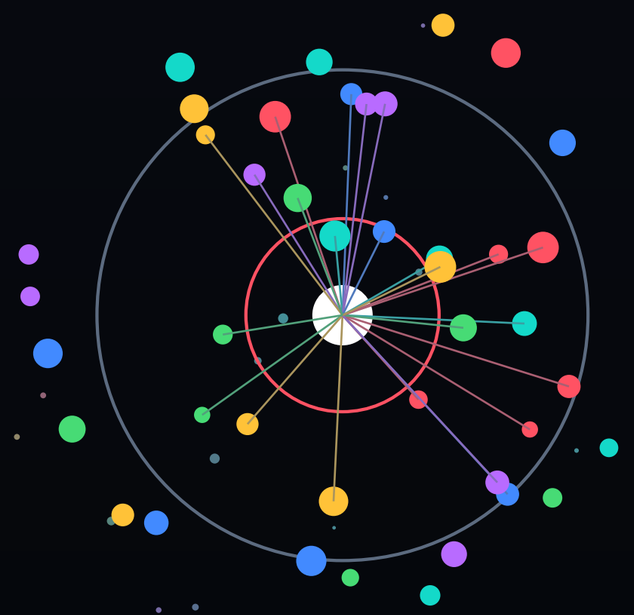


PARALLEL UPDATE

One CUDA thread owns one particle

Each thread reads the old world, gathers the forces on its particle, and writes one new position.

- The work is independent per particle.
- All particles update together.
- The frame becomes one big parallel step.

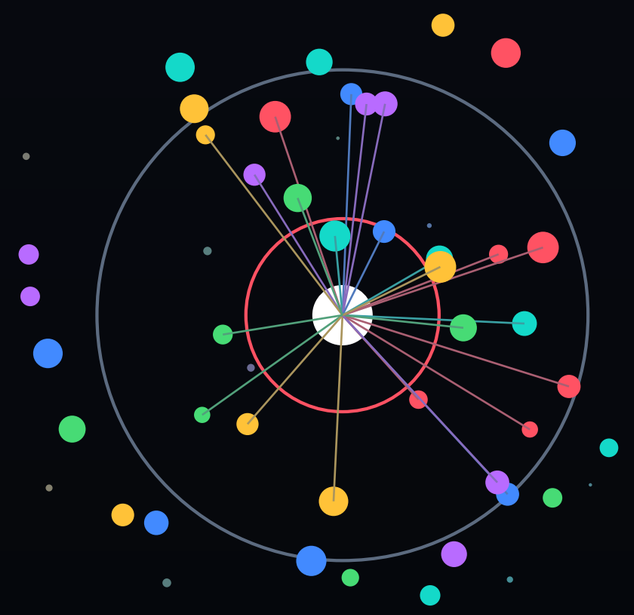


each particle only asks nearby questions

THE EXPENSIVE PART

Each particle checks its neighborhood

The direct version compares particles against each other. It is easy to understand and maps cleanly to CUDA.



each particle only asks nearby questions

near particles matter most

THE RULE LOOKUP

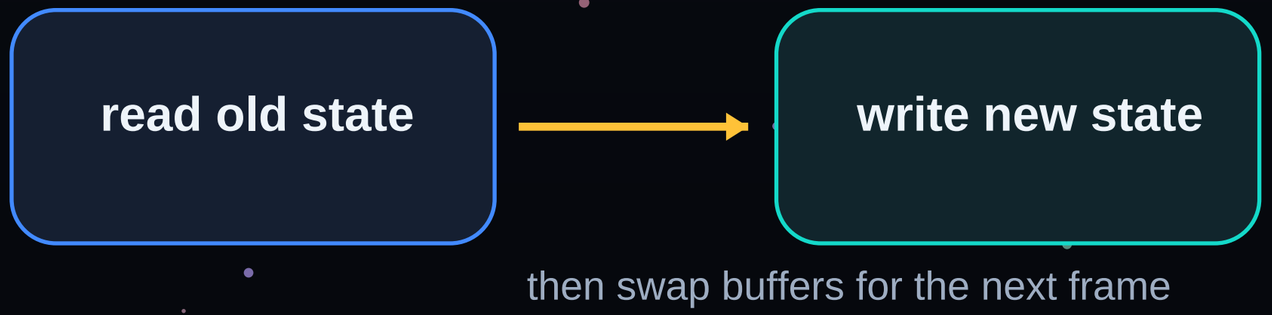
The force matrix stays small

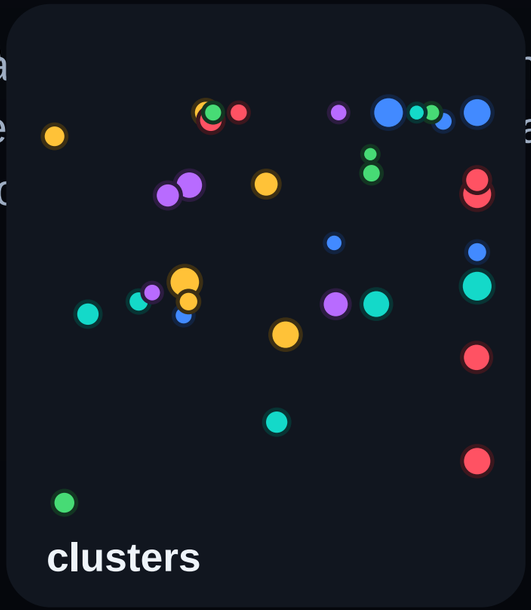
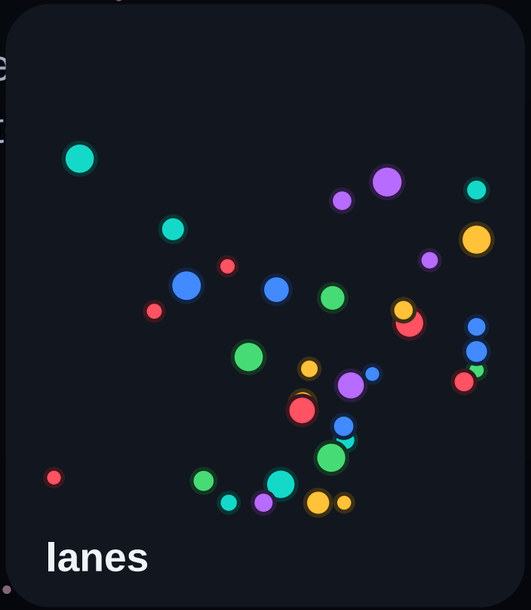
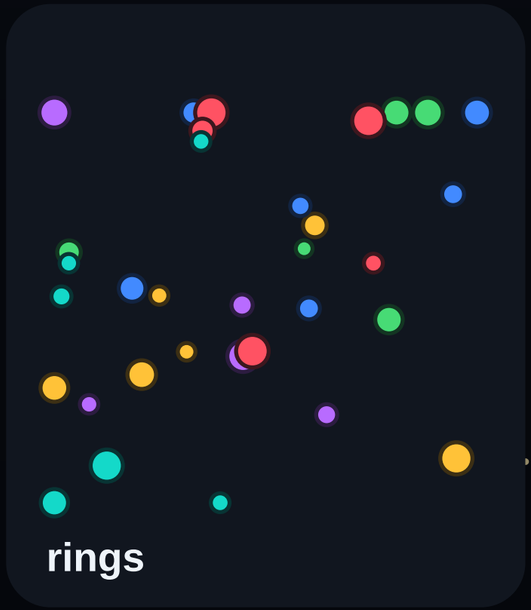
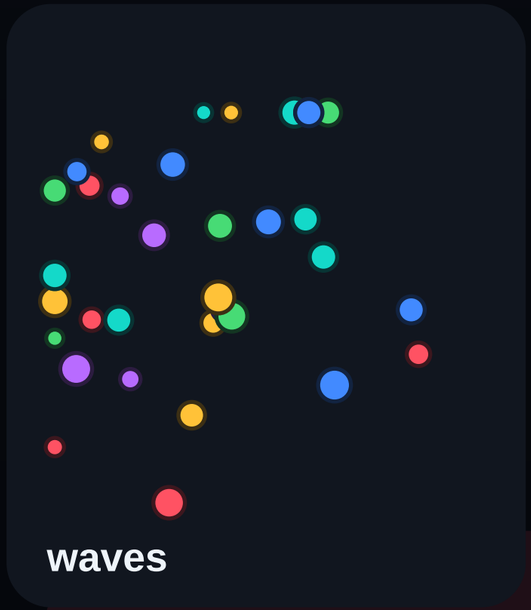
Even with many particles, the rule table is tiny. The GPU mostly spends time applying those rules many times.

CORRECTNESS

Double buffering keeps the frame consistent

Threads read from one copy of the world and write into another. No particle sees a half-updated neighbor.



THE PAYOFF

The GPU turns simple math into live e